Qualcomm Technologies, Inc.

SW6100 Linux Wear OS Performance Technical Overview

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Revision History

Revision	Date	Description
AA	December 2025	Initial release

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Contents

1 Architecture 5

2 QTI Energy Awareness Scheduling Performance Features 8

3 Scheduler Update on SW6100 10

4 Scheduler Features on SW6100 16

5 Touch Boost 29

6 Governor (Walt) 31

7 Core Control (Halt) 36

8 Debugging 42

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Objectives

At the end of this presentation, you will understand the following performance-related aspects for this chipset:

- System architecture
- Scheduler new features and update.

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Section 1

Architecture

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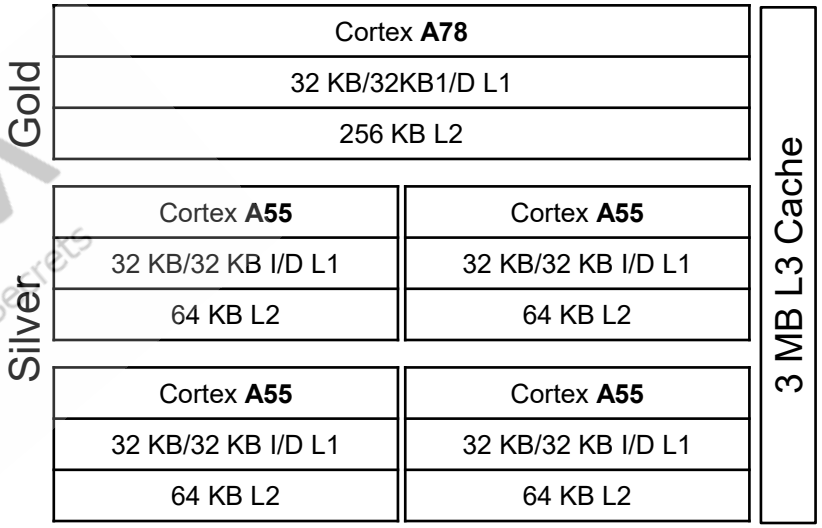
SW6100 Applications Processor

- SW6100 has 1 × Arm® Cortex A78 Gold core:
  - 32 kB instruction cache, 32 kB data cache
  - 256 kB of L2 cache
  - Max frequency: 1.95 GHz
- SW6100 has 4 × Cortex A55 Silver cores:
  - 32 kB instruction cache, 32 kB data cache.
  - 64 kB of L2 cache
  - Max frequency: 2.1 GHz
- All cores share 3 MB of L3 cache.

CPU architecture



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Application Processor Performance Overview

Features	Description
Process	3nm process node
CPU SS	1 × A78 2.1 GHz (Arm Cortex), 256 kB L2 cache, 32 kB L1 cache 4 × A55 1.95 GHz (Arm Cortex), 64 kB L2 cache, 32 kB of dedicated L1 cache 3 MB L3 cache
DDR SS	DDR with 128 KB LLC, LP5 @ 3.2 GHz and LP5.x @ 4.2 GHz support
GPU SS	Qualcomm® Adreno™ 622 v1 graphics

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Section 2

QTI Energy Awareness Scheduling Performance Features

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QTI Energy Awareness Scheduling

- QTI energy awareness scheduling (EAS) is designed to fulfill the market requirements.
- Key changes in the QTI energy awareness scheduling (EAS):
  - It is upgraded to Kernel 6.12
  - Walt support is enabled for 2* cluster CPU architecture on SW6100
    - Silver cluster (APC0)
    - Gold cluster (APC1)

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Section 3

Scheduler Update on SW6100

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WALT Highlight on SW6100

- Task migration:
  - Per cgroup up/down migrate
- Cpufreq governor:
  - Zone based target load

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Task Migration: Per cgroup sched_up/down Migration

- New cgroup-based force migration is introduced, which provides better granularity of task migration control.
- Cgroup support: topapp, foreground, background, and other; and it is per cluster setting.
  - `/proc/sys/walt/clusterX/sched_topapp_updownmigrate`
  - `/proc/sys/walt/clusterX/sched_foreground_updownmigrate`
  - `/proc/sys/walt/clusterX/sched_background_updownmigrate`
  - `/proc/sys/walt/clusterX/sched_other_cgroup_updownmigrate`
- {upmigrate,downmigrate} works as a pair. The first entry is the upmigrate percentage and the second entry is the downmigrate percentage.
- Example:
  - `echo 95 85 > /proc/sys/walt/cluster0/sched_background_updownmigrate`
  - 95: upmigrate threshold; 85: downmigrate threshold. Both are in percentage.
- Perflock is supported
- **Note**:
  - `sched_other_cgroup_updownmigrate` will be the "default" one. That is, if one task is not in `topapp/foreground/background` cgroup. It will be controlled by `sched_`**`other`**`_cgroup_updownmigrate`.
  - Legacy upmigrate/downmigrate are outdated and deprecated..
    - `/proc/sys/walt/sched_downmigrate/sched_upmigrate`
    - `/proc/sys/walt/sched_early_downmigrate/sched_early_upmigrate`

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Sched Upmigrate/Downmigrate tunable

- Provides a threshold to migrate a task to the right cluster to improve UX performance.
- From SW6100, we provide cgroup based task force migration as below table.
- Sched group upmigrate/downmigrate is disabled by default.
- Lower value is performance oriented, and the higher value is power-oriented.

Name	Default value	Unit	Perflock opcode
sched_topapp_updownmigrate	95 85	Percentage	MPCTLV3_SCHED_TA_UPDOWNMIGRATE (0x43C80000)
sched_foreground_updownmigrate	95 85	Percentage	MPCTLV3_SCHED_FG_UPDOWNMIGRATE(0x43C78000)
sched_background_updownmigrate	95 85	Percentage	MPCTLV3_SCHED_BG_UPDOWNMIGRATE (0x43C74000)
sched_other_cgroup_updownmigrate	95 85	Percentage	MPCTLV3_SCHED_OT_UPDOWNMIGRATE (0x43C7C000)
sched_group_upmigrate	**0**		MPCTLV3_SCHED_GROUP_UPMIGRATE(0x40C54000)
sched_group_downmigrate	**0**		MPCTLV3_SCHED_GROUP_DOWNMIGRATE(0x40C58000)
sched_coloc_downmigrate_ns	400000000		MPCTLV3_SCHED_COLOC_DOWNMIGRATE_NS(0x40C8C000)

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NOTE:

- This section will be updated later with the latest clock plan.
- Early up/down migration is deprecated

Waltgov: Zone-based Target Load

- This feature specifies the frequency of zone and the equivalent target load percentage to jump to next frequency.
- Configurable as {freq, percentage} zones.
- Example:
  - `echo 883200 95 1190400 90 > /sys/devices/system/cpu/cpufreq/policy0/walt/zone_max_util_pct`
  - above means next frequency will be range of
    - 1 ~ 883M if load is reaching 95%;
    - 884M ~ 1.19 G if load is reaching 90%
    - Other 1.19 G ~ Fmax if load is reaching 80%
- Perflock supported
- **Note**:
  - Max to 10 zones supported.
  - Target load percentage is range of {80,100}
  - This should be in descending order for each zone.
    - For example: illegal setting: 1190400 95 883200 90
  - Zones and target load (percentage) can change during runtime.
    - `/sys/devices/system/cpu/cpufreq/policy0/walt/zone_max_util_pct`
    - If the user updates the number of zones and then sets the corresponding target load values, any previously configured values for the old zones will be reset.
    - `target_load_thresh/target_load_shift` are outdated and deprecated.
  - Perflock support: `MPCTLV3_ZONE_MAX_UTIL_PCT(0x6081C000)`
    - For example: `0x6081C000,4, 883200, 95,1190400,90`

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Tunable

Name	zone_max_util_pct
Location	`/sys/devices/system/cpu/cpuX/cpufreq/walt/zone_max_util_pct`
Default value	2147483647 80
Description	{freq, percentage} zones
Perflock	`MPCTLV3_ZONE_MAX_UTIL_PCT(0x6081C000)`

- Configurable as {freq, percentage} zones.
- Example : `echo 883200 95 1190400 90 > /sys/devices/system/cpu/cpuX/cpufreq/walt/zone_max_util_pct`
- Perflock support: `MPCTLV3_ZONE_MAX_UTIL_PCT(0x6081C000)`
  - For example: `0x6081C000,4, 883200,95, 1190400,90`

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Section 4

Scheduler Features on SW6100

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Sched Boost (CGROUP Boost)

- Aids groups of tasks to run on performance-friendly cores.
- Variants:
  - Full Throttle boost: All tasks promoted to all CPUs
  - Conservative boost – No force upmigrate is required, only enable early detection to address the latency sensitive case.
  - Restrained boost – No task promotions are made in this variant; distributed load from top app tasks drives CPU frequency.
  - Storage boost – Provides more CPU resources to meet the high I/O throughput case.
- Perflock opcode – `MPCTLV3_SCHED_BOOST(0x40C00000)`
- Sysfs node: `/proc/sys/walt/sched_boost`

Boost type	Boost value	Colocation	Frequency aggregate	Force upmigrate	Early detection	C-state and power cost considerations	Core control*
Full throttle	1	Top app	Yes	All tasks	Yes	No	max_cpus*
Conservative	2	No	No	No	Yes	▪ No – Top app and foreground tasks ▪ Yes – Other tasks	Default
Restrained	3	Top app	Top app	No	No	Yes	Default
Storage boost	4	No	No	No	No	No	max_cpus**
Balance boost	5	Top app	No	No	Yes	Yes	max_cpus*
Off	0	Top app	No	No	No	Yes	Default

NOTE:
* Core CTL configuration may be changed accordingly.
** If max_cpus is limited by some reason, like thermal, sched_boost(1) will not be able to use all the CPUs.

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Early Detect

- Early detection helps to reduce the latency to report the load for some latency-sensitive cases.
  - Only for boost case (boost policy enable)
  - Task wake-up time gap away from last wake-up is longer than 9.5 ms.
- New tunable

Tunable	Comments	Perflock opcode
`sched_ed_boost`	■ `/proc/sys/walt/sched_ed_boost` ■ Percentage; if the default is 10 then it gives 110% boost to load	–

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Storage Boost

- If there is a higher QoS request from a UFS driver, a storage boost is enabled to provide more CPU resources to get better I/O performance.
- Storage boost enables boost for core CTL and unhalt all the possible CPUs.
- During storage boost, the fmax cap is not applied.

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Task-oriented Scheduling

- Apart from the CGROUP-based boost, a smaller granularity schedule policy is also provided to get more precise scheduling accuracy with the following tunable:

  `/proc/sys/walt/sched_task_read_pid` gives the scheduler an ability to read the task ID*

Tunable	Comments	Perflock opcode
`sched_per_task_boost`	`sched_per_task_boost` is to boost the specific task to gold CPU using the PID ▪ Boost to Mid – Disabled and does not work on SW6100 ▪ Boost to Max – Task should run on Gold CPU ▪ Strict Boost to Max – Task should run on Gold, even if Gold gets overutilized.	`MPCTLV3_SCHED_TASK_BOOST(0x40C80000)`
`sched_per_task_boost_period_ms`	Boost duration	–
`sched_init_task_load`	Give task init load when the task is forked.	`MPCTLV3_SCHED_INIT_TASK_LOAD(0x40C18000)`
`sched_low_latency`	Give latency sensitive CFS task a high possibility to be picked to run	`MPCTLV3_SCHED_LOW_LATENCY(0x40CD0000)`
`sched_wake_up_idle`	Wake up task running on an idle CPU	`MPCTLV3_SCHED_WAKE_UP_IDLE(0x40CF0000)`
`task_load_boost*`	To give a specific task load boost (rang -90%~90%)	This information will be updated in a later revision of this document.
`task_reduce_affinity`	To give a specific RT task like SurfaceFlinger to reduce the affinity to avoid power impact. It is not needed on SW6100.	–

NOTE:

- Usage sequence – Read the task ID and then set the per task tunable. Ensure that you disable the setting explicitly or use perflock, which passes the PID.
- *As group migration is disabled by default, the colocated task features are also disabled by default.

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LPM Bias

- Sched-driven Idle mode guidance.
- Big cores usually have large private caches; caches must be flushed when the core enters deep Low-power modes.
- It takes hundreds of microseconds to bring BIG core out of the power collapse.
- Performance bottleneck occurs since BIG cores take a long time to respond and load. The power bottleneck (increased cache flushing activity) occurs if BIG cores are constantly entering/exiting deep LPMs.
- Sched code tracks various parameters on the CPU and guides the idle CPU governor to hold CPUs in shallow Low-power modes.

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Busy Hysteresis Tunables

Tunable	Comment	Unit	Perflock
sched_busy_hysteresis_enable_cpus	CPU mask contributes to the busy hysteresis	CPU mask	MPCTLV3_SCHED_BUSY_HYSTERSIS_CPU_MASK
sched_busy_hyst_ns	Default CPU busy hysteresis time	ns	MPCTLV3_SCHED_COLOC_BIAS_HYST
sched_coloc_busy_hysteresis_enable_cpus*	Colocate CPU mask contributes to the busy hysteresis	Colocate CPU mask	MPCTLV3_SCHED_BUSY_HYSTERESIS_ENABLE_COLOC_CPUS
sched_coloc_busy_hyst_max_ms*	Maximum colocate time; if the RTB time is longer than the maximum, update the hysteresis directly.	ms	–
sched_coloc_busy_hyst_cpu_busy_pct*	Percentage of per CPU cap	Percentage	MPCTLV3_SCHED_COLOC_BUZY_HYST_CPU_PCT
sched_coloc_busy_hyst_cpu_ns*	Per CPU hysteresis busy time	ns	MPCTLV3_SCHED_COLOC_BUZY_HYST_CPU_NS
sched_util_busy_hyst_cpu_util	Util of per CPU	–	MPCTLV3_SCHED_UTIL_BUSY_HYST_CPU_NS_BIG

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NOTE: * As group migration is disabled by default, the colocated task features are disabled by default.

Task Filter

Name	sched_min_task_util_for_boost
Location	`/proc/sys/walt/sched_min_task_util_for_boost`
Default value	51
Description	Unit in percentage
Perflock opcode	`MPCTLV3_SCHED_MIN_TASK_UTIL_FOR_BOOST(0x40C7C000)`

Name	sched_min_task_util_for_colocation
Location	`/proc/sys/walt/sched_min_task_util_for_colocation`
Default value	35
Description	Unit in percentage
Perflock opcode	`MPCTLV3_SCHED_MIN_TASK_UTIL_FOR_COLOCATION(0x40C78000)`

- These `sched_min_task_util_for_boost` and `sched_min_task_util_for_colocation` tunables determine if:
  - A task is in colocation
  - The task with demand is **lower** than the `/proc/sys/walt/sched_min_task_util_for_colocation` threshold, then it stays on the Silver cluster.
  - The task with demand is **higher** than the `/proc/sys/walt/sched_min_task_util_for_boost` threshold, it would be a boost on to the Gold clusters.

Name	sched_task_unfilter_period
Location	`/proc/sys/walt/sched_task_unfilter_period`
Default value	100000000
Description	Unit in ns
Perflock opcode	`MPCTLV3_SCHED_TASK_UNFILTER_PERIOD(0x40CB8000)`

- The `sched_task_unfilter_period` tunable monitors the task filter valid time. The default time is 100 ms.
- Lower value is performance-oriented and the higher value is power-oriented.
- `sched_min_task_util_for_uclamp` for migration is disabled on SW6100.

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sched_many_wakeup_threshold

- In a few use cases, a task wakes the other tasks back and forth, which crowds a single cluster. The `sched_many_wakeup_threshold` tunable provides the threshold to avoid such scenarios.
- If the wake-up task count on a specific CPU is greater than the threshold, the next CPU is skipped.
- The default value is 1000.
- Perflock opcode: `MPCTLV3_SCHED_MANY_WAKEUP_THRESHOLD (0x40C9C000)`

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sched_user_hint

- The change attempts to boost the top-app hosting cluster frequency.
- The sysnode gives the top-app load to trigger the frequency boost:

$$load = \frac{load \ * sched_user_hint}{100}$$

- This hint rests to 0 after 1 sec.
- Hint range is 0 to 1000.
- Perflock opcode: `MPCTLV3_SCHED_USER_HINT(0x40C84000)`

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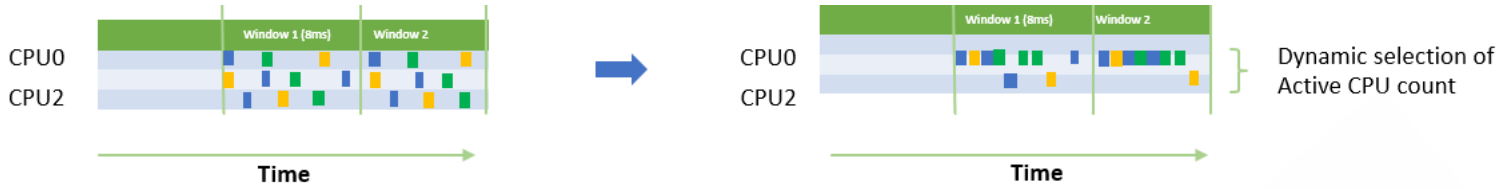
Task Packing Control



- Packing tasks to the same CPU benefits power but may impact performance. There must be a balance between performance and power for some specific cases.
- Here are some conditions that can control the packing fast path:
  - Packing tasks in the same cluster, if possible.
  - Packing happens only on unhalted CPUs.
  - Packing is not possible:
    - If `sched_idle_enough` = 0 (false), the fast path packing is disabled
    - If the current cluster load is over `sched_cluster_util_thres_pct`
    - High load CPU (higher than sched_idle_enough) does not accommodate packing
    - High demand task (higher than sched_idle_enough) is packing
    - When the average load is dropping
    - Packing is not possible on partial halt CPU
    - For any other use cases, packing tasks can happen to save power:

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Tunable	Comment	Unit	
`sched_idle_enough`	Threshold to decide possible packing: ▪ High demand task ▪ High util CPU	Task demand	`MPCTLV3_SCHED_IDLE_ENOUGH`
`sched_idle_enough_clust`	Threshold to decide possible packing: ▪ High util cluster ▪ High util CPU	CPU util	`MPCTLV3_SCHED_IDLE_ENOUGH_CLUST_BIG` `MPCTLV3_SCHED_IDLE_ENOUGH_CLUST_LITTLE`
`sched_cluster_util_thres_pct`	Threshold to decide cluster is high load or not	Percentage	`MPCTLV3_SCHED_CLUSTER_UTIL_THRES_PCT`
`sched_cluster_util_thres_pct_clust`	Threshold (per cluster) to decide which cluster is high load or not	–	`MPCTLV3_SCHED_CLUSTER_UTIL_THRES_PCT`

Smart Frequency Management

- Workload with low IPC will see less performance improvement from using higher frequencies but can still burn high power.
- Frequency ceiling management
  - A smart frequency algorithm considers both the IPC and other workload characteristics to determine the appropriate frequency ceiling for the CPU.

  - Algorithm re-evaluates the workload every sched tick (4 ms) to determine the correct frequency ceiling level on each cluster.
    - For example, in a cluster with two CPU cores, the 1st core ceiling is fixed to 1.8 GHz, and the second core ceiling is fixed to 2 GHz, then max ceiling is chosen for that cluster.

- Workload based frequency ceiling selection input
  - Frequency ceiling determination also considers other aspects beyond IPC, such as # misfits in the system, wake up count, as well as sustained high-load conditions.
  - Higher frequency can still be required even with lower IPC for scenarios with heavy load and low-performance tolerance. In these cases, the IPC-based frequency ceiling can be overwritten.

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Smart Frequency Management - Tunable

- **IPC based frequency ceiling selection input**
  - IPC zones are defined to represent workload efficiency, and each IPC zone has a corresponding frequency ceiling.
  - The highest IPC among the CPUs in cluster will set the frequency ceiling based on preset thresholds.
- **Ways to uncap the IPC based frequency ceiling:**
  - use `scaling_min_freq` to boost
  - use full throttle boost
  - update `ipc_reason_config[level].freq_allowed`
- Debug:
  - `trace_ipc_update`
  - `trace_ipc_freq`
  - `trace_sched_freq_uncap`
  - `trace_waltgov_util_update`
  - `trace_waltgov_next_freq`

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Tunable	Default value	Unit/Parameters	Perflock
`/proc/sys/walt/clusterX/smart_freq/ipc_freq_levels`	For example: `2400000 2400000 2700000 3000000 2147483647`	Frequency	`MPCTLV3_SCHED_IPC_LEVELS_BIG(0x63C50000)`
`/proc/sys/walt/clusterX/smart_freq/legacy_freq_levels`	`0:10:2147483647:0:0`	{cluster,Index, frequency, Hysteresis , enable/Disable}	`MPCTLV3_SCHED_LEGACY_FREQ_LEVELS(0x63C5C000)`

`/proc/sys/walt/cluster*/smart_freq/sched_smart_freq_dump_ipc_reason`	`0:0:0:2147483647:0:0`
	`0:1:0:2147483647:0:0`
	`0:2:0:2147483647:0:0`
	`0:3:0:2147483647:0:0`
	`0:4:0:2147483647:0:0`
	`1:0:0:2147483647:0:0`
		{cluster,Index,IPC Value, frequency, Hysteresis , enable/Disable}	NA

Section 5

Touch Boost

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About Touch Boost

- Touch Boost feature gives a system boost when the touch event is received. This provides better UX performance for the improved input latency.
- To follow the GKI change, the Touch Boost-related nodes should be moved to the same folder as Walt.

Path	Kernel 5.4	Kernel 5.10 and later versions
Code	`techpack/sched/cpu-boost.c`	`kernel/sched/walt/input-boost.c`
Tunable	`/sys/devices/system/cpu/cpu_boost/`	`/proc/sys/walt/input_boost/`

- Tunable:

Tunable	Comments	Format
`input_boost_freq`	Boost frequency	{cpu:freq} Example: 0:960000 1:0 2:0 3:0 4:0 5:0
`input_boost_ms`	Boost duration	In ms

- Lower (short) value is power-oriented and the higher (longer) value is performance-oriented.
- Perflock opcode `MPCTLV3_INPUT_BOOST_RESET(0x42C08000)` support is to change the input boost frequency for some specific cases.

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Section 6

Governor (Walt)

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About Walt Governor

- SchedUtil Governor is no longer used after Kernel 5.10. Instead, Walt Governor is used for the CPU frequency guidance.
- Walt works with the EAS to change the CPU frequency. It also supports `fast_switch` by default.
- Cpufreq driver follows GKI upstream SCMI cpufreq driver, which is specification-compliant.
- `fast_switch` does the following:
  - Facilitates frequency switching from the scheduler's context without spawning the `waltgov` thread.
  - Reduces the latency of the switching frequency.

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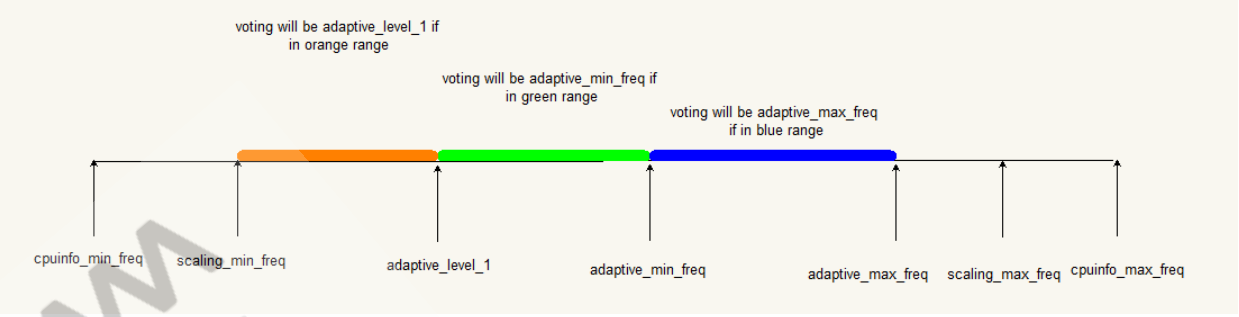
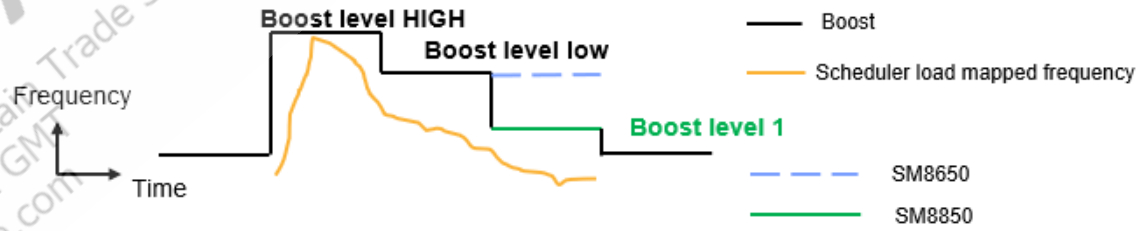
Walt Tunable – adaptive_freq (CPU Load Aware Boost)

- Added adaptive_level_1 to minimize possible power impact from loss of Silver cores during lightly-loaded portion of Launch
- These tunables give frequency bucket in some use cases:
  - If the vote frequency < `adaptive_level_1`, target frequency is `adaptive_level_1`.
  - If the vote frequency is between `adaptive_level_1` and `adaptive_low_freq`, the target frequency is `adaptive_low_freq`.
  - If the vote frequency is between `adaptive_low_freq` and `adaptive_high_freq`, the target frequency is `adaptive_high_freq`.
- This avoids governor voting latency when the load is not high enough.
- Lower value is power-oriented and higher value is performance-oriented.





Name	adaptive_high_freq
Location	`/sys/devices/system/cpu/cpuX/cpufreq/walt/adaptive_high_freq`
Default value	0 (This information will be updated in a future revision of this document)
Description	Unit in Hz
Perflock code	`MPCTLV3_WALT_ADAPTIVE_HIGH_FREQ_BIG(0x4080C000)`

Name	adaptive_low_freq
Location	`/sys/devices/system/cpu/cpuX/cpufreq/walt/adaptive_low_freq`
Default value	0 (This information will be updated in a future revision of this document)
Description	Unit in Hz
Perflock code	`MPCTLV3_WALT_ADAPTIVE_LOW_FREQ_BIG(0x40808000)`

Name	adaptive_level_1
Location	`/sys/devices/system/cpu/cpuX/cpufreq/walt/adaptive_level_1`
Default value	0 (This information will be updated in a future revision of this document)
Description	Unit in Hz
Perflock code	`MPCTLV3_WALT_ADAPTIVE_LEVEL_1_FREQ_BIG(0x40814000`

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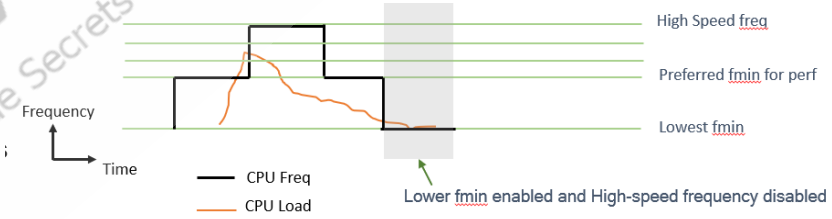
Walt Tunable - High-speed

Name	hispeed_freq
Location	`/sys/devices/system/cpu/cpuX/cpufreq/walt/hispeed_freq`
Default value	1036800/1440000
Description	▪ **Frequency** bump to reach hispeed_load ▪ Lower value is power-oriented, higher value is performance-oriented
Perflock	▪ `MPCTLV3_SCHEDUTIL_HISPEED_FREQ_CLUSTER_BIG(0x4143C000)`

Name	hispeed_load
Location	`/sys/devices/system/cpu/cpuX/cpufreq/walt/hispeed_load`
Default value	90/90
Description	▪ **Percentage** load to bump to hispeed_freq ▪ Lower value is power-oriented, higher value is performance-oriented
Perflock	▪ `MPCTLV3_SCHEDUTIL_HISPEED_LOAD_CLUSTER_BIG(0x41440000)`

Name	hispeed_cond_freq
Location	`/sys/devices/system/cpu/cpuX/cpufreq/walt/hispeed_cond_freq`
Default value	0
Description	▪ **Frequency** threshold ▪ If avg_load is dropping below the threshold, hiload can be break
Perflock	▪ `MPCTLV3_WALT_HISPEED_COND_FREQ_BIG(0x40818000)`



- If the average load reaches `hispeed_load`, the frequency bumps up to `hispeed_freq`
- Reduces the latency of CPU frequency ramping.
- Tunable pairs match the cluster numbers.
- [New] hispeed_cond_freq will give the change that break the hiload condition if the avg_load is dropping
- Lower value is power-oriented and higher value is performance-oriented.
  Load will be avg – top app load

NOTE: * This section would be updated later with clock plan details.

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Walt Tunable – rtg_boost

Name	rtg_boost_freq
Location	`/sys/devices/system/cpu/cpuX/cpufreq/walt/rtg_boost_freq`
Default value	249600/960000
Description	Frequency
Perflock opcode	`MPCTLV3_SCHEDUTIL_RTG_BOOST_FREQ_CLUSTER_BIG(0x4144C000)`

- This tunable can give higher boost input for specific clusters when the target CPU frequency for the colocated group tasks is selected.
- When colocation is active, that is, top application is preferred and the cluster is Gold, the CPU frequency is pegged at `rtg_boost_frequency` minimum. At present, it is used only for the Silver cluster so that filtered tasks from the Gold cluster can use the higher frequency.
- Lower value is power-oriented and higher value is performance-oriented.

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NOTE: Ensure that you enable the group upmigrate at the same time when you set rtg_boost. * This section would be updated later with the clock plan details.

Section 7

Core Control (Halt)

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Introduction

- No partial halt on SW6100.
- Node: `/sys/devices/system/cpu/cpuX/core_ctl/`

Parameter	Silver	Gold
`core_ctl enable`	1	1
`min_cpus`	1	0
`Max_cpus`	4	1
`not_preferred`	0 1 1 1	0

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Halt Tunable

Name	busy_down_thres
Location	`/sys/devices/system/cpu/cpuX/core_ctl/busy_down_thres`
Default value	30 30 30 30 (Silver)/30 (Gold)
Description	Unit in percentage

Name	busy_up_thres
Location	`/sys/devices/system/cpu/cpuX/core_ctl/busy_up_thres`
Default value	60 60 60 60 (Silver)/60 (Gold)
Description	Unit in percentage

- If the load of the CPU is under the busy down threshold, the CPU is marked as not busy.
- If the load is above the busy up threshold, the CPU is marked as busy and may need more CPU online.
- Lower value is performance-oriented and higher value is power-oriented.

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Halt Tunable (cont.)

Name	min_cpus
Location	`/sys/devices/system/cpu/cpuX/core_ctl/min_cpus`
Default value	1 (Silver)/ 0 (Gold)
Description	Core number
Perflock	▪ `MPCTLV3_MIN_ONLINE_CPU_CLUSTER_BIG(0x41000000)`

Name	max_cpus
Location	`/sys/devices/system/cpu/cpuX/core_ctl/max_cpus`
Default value	4 (Silver)/1 (Gold)
Description	Core number
Perflock	▪ `MPCTLV3_MAX_ONLINE_CPU_CLUSTER_BIG(0x41004000)`

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Halt Tunable (cont.)

Name	Task_thres
Location	`/sys/devices/system/cpu/cpuX/core_ctl/task_thres`
Default value	4(Silver)/1(Gold)
Description	Heavy task number

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NOTE: If the tasks on the cluster are more than the task_thres, all the cores on the cluster are unisolated.

offline_delay_ms

Name	offline_delay_ms
Location	`/sys/devices/system/cpu/cpuX/core_ctl/offline_delay_ms`
Default value	1000/1000 (ms)
Description	This tunable allows to change the CPU halt/unhalt behavior.

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Section 8

Debugging

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EAS Trace Points in Kernel 6.12

- All the upstream sched traces are enabled using "`sched:*`" name

  `adb shell "echo sched:* >> /sys/kernel/tracing/set_event”`

  - Example:
    - `adb shell "echo sched:sched_waking >> /sys/kernel/tracing/set_event”`
    - `adb shell "echo sched:sched_wakeup >> /sys/kernel/tracing/set_event”`
- Walt traces are enabled by using “`schedwalt:*`” name

  `adb shell "echo schedwalt:* >> /sys/kernel/tracing/set_event”`

  - Example:
    - `adb shell "echo schedwalt:sched_find_best_target >> /sys/kernel/tracing/set_event”`
    - `adb shell "echo schedwalt:sched_task_util >> /sys/kernel/tracing/set_event”`

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Acronyms and Terms

Acronym or term	Definition
DCVS	Dynamic clock and voltage scaling
EAS	Energy aware scheduling

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Questions?

For additional information or to submit technical questions, go to: https://www.qualcomm.com/support

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